# Ascent Performance of the Rockot/Breeze-KM Launch Vehicle from Plesetsk

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**Abstract.** A three-degree-of-freedom ascent model of Rockot/Breeze-KM is calibrated against the reference mission of the User's Guide [1]: 1,570 kg to a 650 km circular orbit at 86.58°. Stage-1 and stage-2 burnout speeds agree within 0.2 % and 0.5 %. Maximum dynamic pressure is 64.8 kPa and peak axial load 7.2 g. The upper stage inserts into a 643 × 666 km orbit at 86.34° with 235 kg of propellant remaining.

## 1 Vehicle

Engine, specific impulse and burn-time data are from [1], dry masses from [2]. Stage-1 and stage-2 propellant loads are fitted (Section 4); thrust follows from load, burn time and specific impulse. Lift-off mass is 117.0 t and lift-off thrust-to-weight ratio 1.75.

**Table 1.** Stage parameters. Ideal $\Delta v$ totals 10,061 m/s (fairing carried to stage-2 burnout).

| | Stage 1 | Stage 2 | Breeze-KM |
|---|---|---|---|
| Engines | RD-0233 ×3, RD-0234 | RD-0235, RD-0236 ×4 | S5.98M |
| Thrust, vacuum (kN) | 2,178 | 248 | 20 |
| $I_{sp}$, vacuum (s) | 310.0 | 320.0 | 325.5 |
| $I_{sp}$, sea level (s) | 285.0 | – | – |
| Burn time (s) | 121 | 183 | ≤ 794 |
| Propellant (kg) | 86,691 | 14,475 | 4,975 |
| Dry mass (kg) | 5,695 | 1,485 | 1,320 |
| Ideal $\Delta v$, vacuum (m/s) | 4,109 | 2,790 | 3,161 |

## 2 Mission performance

**Table 2.** Reference mission, Plesetsk (62.8° N, 40.3° E).

| | Model | User's Guide [1] |
|---|---|---|
| Payload (kg) | 1,570 | 1,570 |
| Lift-off mass (t) | 117.0 | 107 |
| Max. dynamic pressure (kPa) | 64.8 at 51 s | 61.3 at 50 s |
| Peak axial load, stage 1 (g) | 7.2 | 6.2–8.0 |
| Peak axial load, stage 2 (g) | 2.7 | ≤ 3.0 |
| Peak axial load, Breeze-KM (g) | 0.64 | – |
| Transfer orbit (km) | 160 × 635 | 160 × 635 |
| Final orbit (km) | 643 × 666 | 650 × 650 |
| Inclination (deg) | 86.34 | 86.58 |
| Lift-off to transfer orbit (s) | 1,038 | 882 |
| Breeze-KM propellant remaining (kg) | 235 | – |

## 3 Trajectory

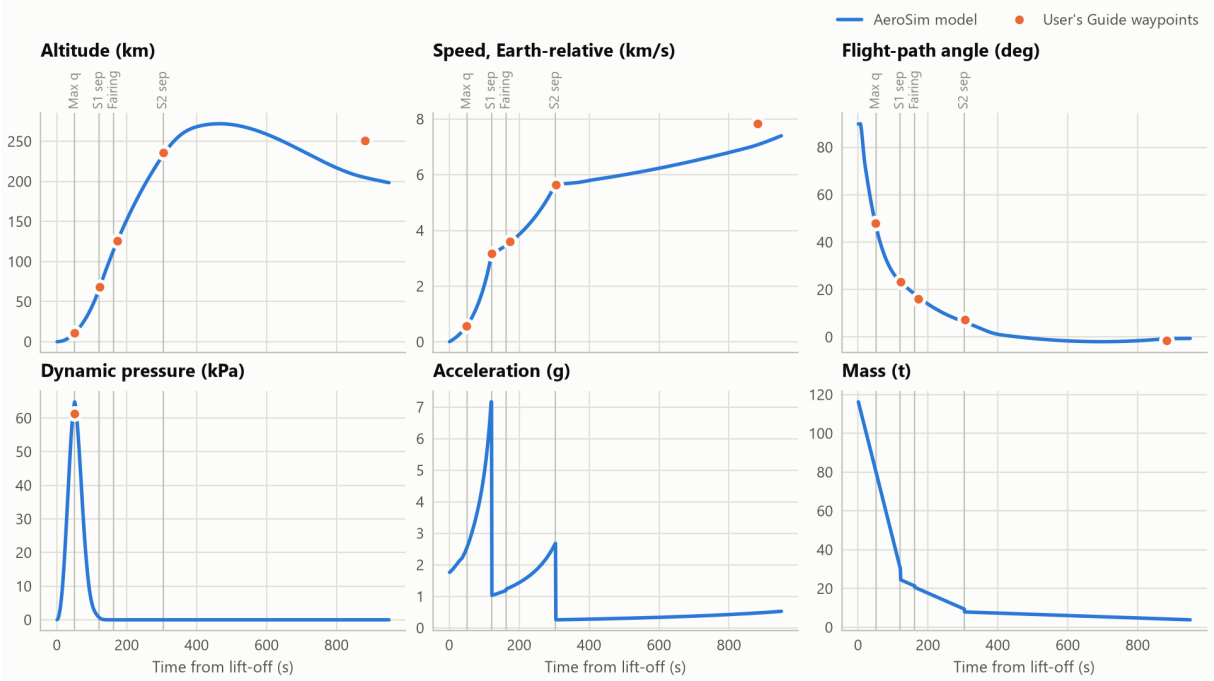



**Figure 1.** Ascent to the end of Breeze-KM burn 1. Line: model; points: waypoints of [1, fig. 3-6]. Vertical lines mark max. *q*, stage-1 separation, fairing jettison and stage-2 separation.

**Table 3.** State at flight events: model against [1, fig. 3-6]. *v*: Earth-relative speed; γ: flight-path angle.

| Event | *t* (s) | | *h* (km) | | *v* (m/s) | | γ (deg) | |
|---|---|---|---|---|---|---|---|---|
| | Model | Guide | Model | Guide | Model | Guide | Model | Guide |
| Max. *q* | 51 | 50 | 10.9 | 10.7 | 591 | 567 | 46.9 | 47.9 |
| Stage-1 sep. | 121 | 122 | 65.9 | 68.3 | 3,156 | 3,163 | 23.3 | 23.3 |
| Fairing jettison | 162 | 173 | 114.2 | 125.6 | 3,464 | 3,599 | 18.0 | 16.0 |
| Stage-2 sep. | 304 | 305 | 233.0 | 235.7 | 5,663 | 5,634 | 6.5 | 7.3 |
| Burn-1 cutoff | 1,038 | 882 | 189.9 | 250.7 | 7,902 | 7,832 | −0.6 | −1.5 |

## 4 Calibration and limitations

Stage-1 and stage-2 propellant loads, pitch-kick angle, stage-2 pitch rate and the Breeze-KM climb-rate law were fitted by least squares to ten waypoints of [1]; eight are met within tolerance. The two misses, lift-off mass (+9.3 %) and burn-1 cutoff (+156 s), indicate that the model needs more energy than the flight vehicle, consistent with its simplified insertion steering and generic drag curve. The model is a point mass with constant thrust per stage, $J_2$ gravity and an exponential upper atmosphere [3].

## References


[1] Eurockot Launch Services, *Rockot User's Guide*, EHB0003, Issue 5, Rev. 0, 2011.

[2] M. Wade, "Rokot," *Encyclopedia Astronautica*, astronautix.com.

[3] D. A. Vallado, *Fundamentals of Astrodynamics and Applications*, table 8-4.